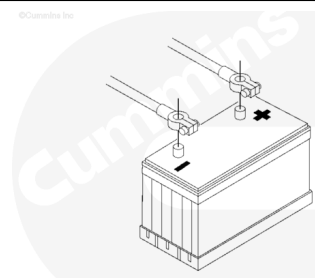


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

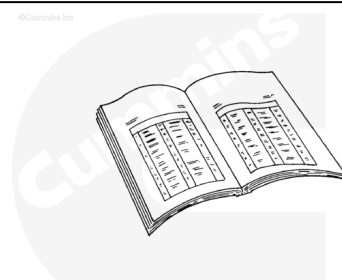
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

- Disconnect the batteries.

- Remove the drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- Remove the vibration damper/crankshaft pulley. If equipped with a viscous damper refer to Procedure 001-052 (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>). If equipped with a rubber damper refer to Procedure 001-051 (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>). If equipped with crankshaft pulley only, refer to Procedure 001-022 (</qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html>)
- Remove the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html>).



ck800wa

Initial Check

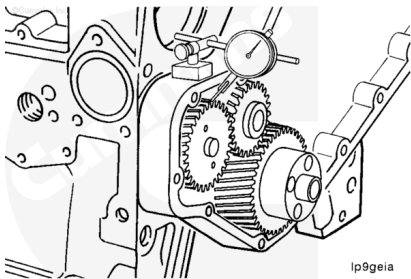
Note : The oil pump gear backlash limits for rear gear train engines and front gear train engines are different. Rear gear train engine use a lubricating oil pump with

straight cut gears and front gear train engines use a lubricating oil pump with helical cut gears.

Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, measure the lubricating pump driven gear backlash.

mm		in
0.170	MIN	0.007
0.300	MAX	0.012

mm		in
0.076	MIN	0.003
0.330	MAX	0.013



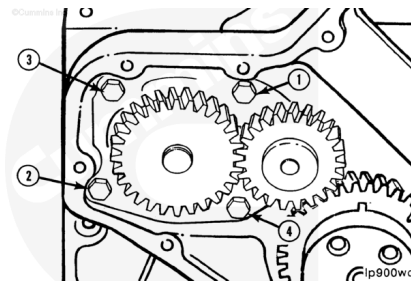
If the backlash is out of specification, the lubricating oil pump **must** be replaced.

Note : If the adjoining gear moves when you measure the backlash, the reading will be incorrect.

Remove

Remove the four mounting capscrews (1, 2, 3, and 4).

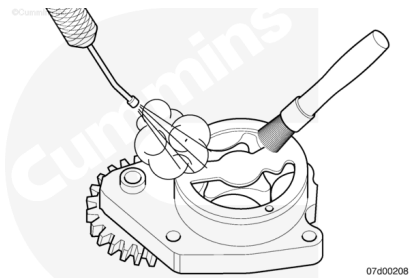
Remove the lubricating oil pump from the bore in the cylinder block.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

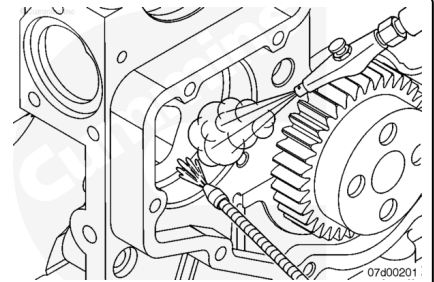
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the lubricating oil pump with solvent.

Dry with compressed air.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

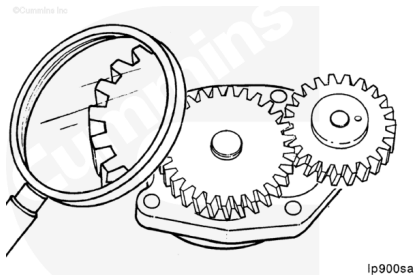
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

With the pump removed use solvents to clean the cylinder block cavity.

Dry with compressed air.

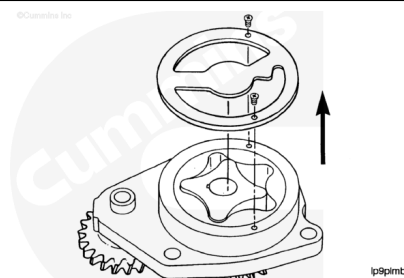
Note : Do not allow solvents to enter the lubricating oil system when cleaning the cylinder block cavity.

Inspect the lubricating oil pump gears for chips, cracks, or excessive wear.

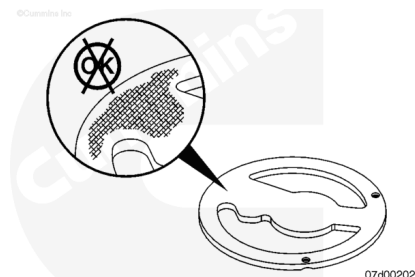


If debris has been expected to have gone through the lubricating oil pump, the rear cover plate can be removed to inspect for internal damage.

Note : The lubricating oil pump is serviced/replaced as an assembly. No internal parts are available for rebuild/replacement.

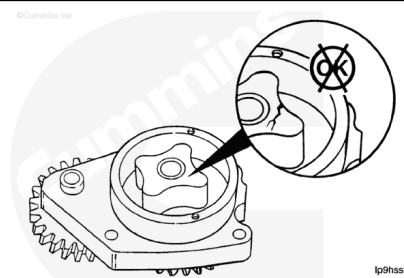


Inspect the rear cover plate for scoring or grooves. If the cover is scored or grooved too badly, the lubricating oil pump **must** be replaced.



Inspect the lubricating oil pump housing and gerotor drive for damage and excessive wear.

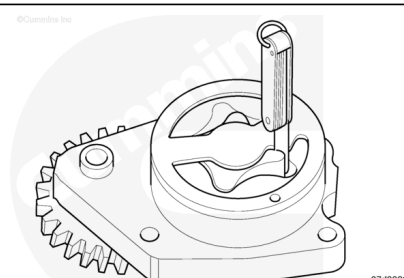
If the lubricating oil pump is damaged, the lubricating oil pump **must** be replaced.



Measure

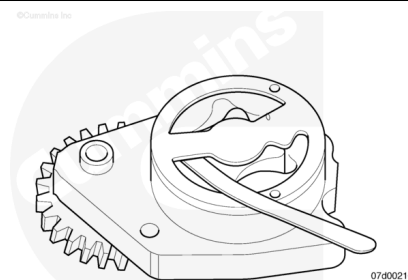
Measure the tip clearance.

mm		in
0.178	MAX	0.007



Measure the clearance of the gerotor drive/gerotor planetary to port plate.

mm		in
0.127	MAX	0.005

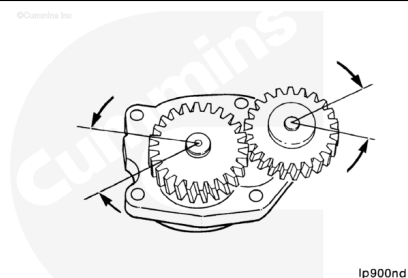


Note : The oil pump gear backlash limits for rear gear train engines and front gear train engines are different. Rear gear train engine use a lubricating oil pump with straight cut gears and front gear train engines use a lubricating oil pump with helical cut gears.

Measure the gear backlash.

mm		in
0.170	MIN	0.007
0.300	MAX	0.012

mm		in
0.076	MIN	0.003
0.330	MAX	0.013

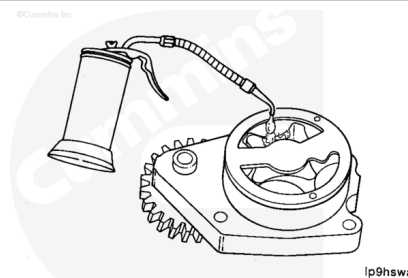


Install

⚠ CAUTION ⚠

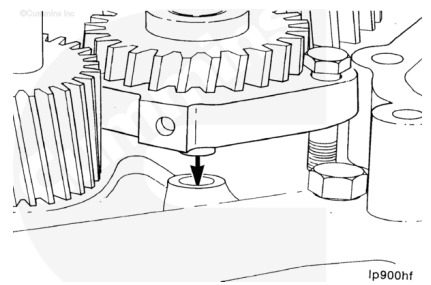
Failure to fill the pump with oil during installation can result in a slow prime at initial engine start-up, resulting in severe engine damage.

Lubricate the lubricating oil pump with clean 15W-40 engine oil.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the idler gear pin is installed in the locating bore in the cylinder block.



Install the lubricating oil pump.

Note : If installing a new lubricating oil pump ensure that the pump is correct for your engine. For example, lubricating oil pumps for 4 and 6 cylinder engines are physically interchangeable, but have different flow characteristics.

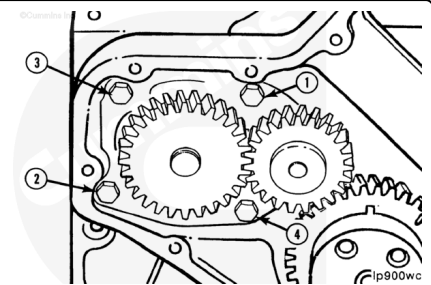
Using the sequence shown, torque the capscrews in the following sequence:

Initial Torque

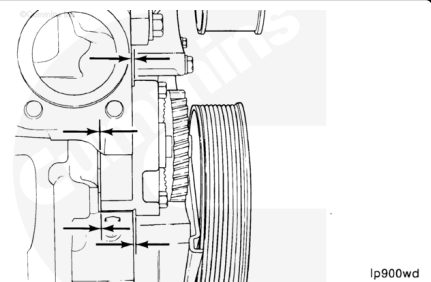
Torque Value: 8 n•m [70 in-lb]

Final Torque

Torque Value: 24 n•m [212 in-lb]



Note : The back plate on the pump seats against the bottom of the bore in the cylinder block. When the lubricating oil pump is correctly installed, the flange on the lubricating oil pump will not touch the cylinder block.

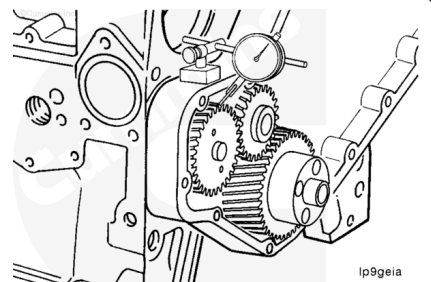


Note : Be sure to check if gear backlash is correct when installing a new lubricating oil pump.

Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, measure the lubricating pump driven gear backlash.

Note : The oil pump driven gear backlash limits for rear gear train engines and front gear train engines are different. Rear gear train engine use a lubricating oil pump with straight cut gears and front gear train engines use a lubricating oil pump with helical cut gears.

Note : If the adjoining gear moves when you measure the backlash, the reading will be incorrect.



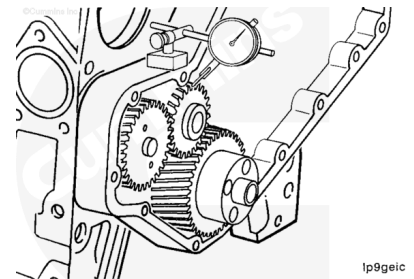
	mm		in

Rear Gear Train	0.300	MIN	0.011
	0.500	MAX	0.019
Front Gear Train	0.076	MIN	0.003
	0.330	MAX	0.013

Note : If you are reinstalling a used pump and have already measured the backlash you do not need to complete this step.

Note : Be sure to check if gear backlash is correct when installing a new lubricating oil pump.

Use gauge, Part Number 3824564, and magnetic base, Part Number 3377399, measure the lubricating pump idler gear backlash.



Note : The oil pump idler gear backlash limits for rear gear train engines and front gear train engines are different. Rear gear train engine use a lubricating oil pump with straight cut gears and front gear train engines use a lubricating oil pump with helical cut gears.

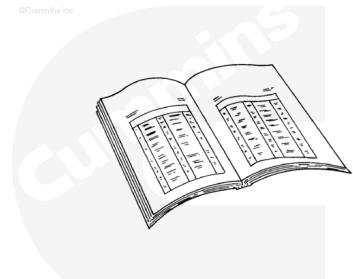
Note : If the adjoining gear moves when you measure the backlash, the reading will be incorrect.

	mm		in
Rear Gear Train	0.150	MIN	0.005
	0.250	MAX	0.009
Front Gear Train	0.076	MIN	0.003
	0.330	MAX	0.013

Note : If you are reinstalling a used pump and have already measured the backlash you do not need to complete this step.

Finishing Steps

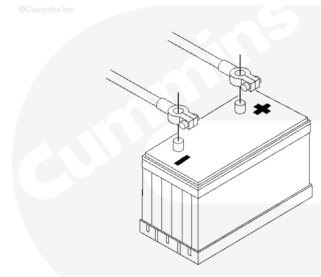
- Install the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/40/40-001-031-tr.html)
- Install the vibration damper/crankshaft pulley. If equipped with a viscous damper, refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html). If equipped with a rubber damper, refer to Procedure 001-051 (/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html). If equipped with a crankshaft pulley only, refer to Procedure 001-022 (/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)
- Install the belt. Refer to Procedure 008-002 (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html).



ck800wa

⚠ WARNING ⚠

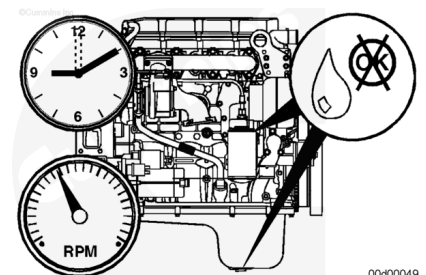
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

- Connect batteries.

- Operate the engine and check for leaks.



00300049